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We are separating a mixture of ethane (C2), propane (C3), n-butane (C4), and n-pentane (C5) in a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. The feed flow rate is 1000 kmol/h. The feed is a saturated liquid. Feed is 10 mol% ethane, 30 mol% propane, 50 mol% n-butane, and 10 mol% n-pentane. K values are $K_{C2}=4.6$, $K_{C3}=1.37$, $K_{C4}=0.35$, and $K_{C5}=0.10$. A 99.5% recovery of propane is desired in the distillate. A 99.0% recovery of n-butane is desired in the bottoms.

ethane : LNK ✓

propane : LK ✓

n-butane : HK ✓

n-pentane : HNK ✓

- 1) (2pts) Identify each component such as LNK, LK, NK, HK, or HNK

- 2) (2pts) Write down fractional recovery (e.g., $FR_{i,D} = x_{i,D} \cdot D / F \cdot z_i$) of ethane, propane, n-butane, and n-pentane both in distillate and bottom

	Distillate	Bottom
Ethane	$FR_{C2,dist} = 1$	$FR_{C2,bot} = 0$
Propane LK	$FR_{C3,dist} = 0.995$	$FR_{C3,bot} = 0.005$
Butane HK	$FR_{C4,dist} = 0.01$	$FR_{C4,bot} = 0.99$
Pentane	$FR_{C5,dist} = 0$	$FR_{C5,bot} = 1$

- 3) (3pts) Calculate the distillate flow rate (D).

$$D = \sum D x_{i,dist} = \sum FR_{i,dist} D x_{i,dist} = F z_{C2} + 0.995 F z_{C3} + 0.01 F z_{C4} + 0 F z_{C5}$$

$$= 1000 (0.1 + 0.995 \times 0.3 + 0.01 \times 0.5)$$

$$= 403.5 \text{ kmol/h}$$

- 4) (3pts) Find N_{min} from the Fenske equation (hint: use the relationship between α and K value)

$$N_{min} = \frac{\ln \left[\frac{(FR_A)_{dist} (FR_B)_{bot}}{[1 - (FR_A)_{dist}] [1 - (FR_B)_{bot}]} \right]}{\ln \alpha_{AB}}$$

where, $(FR_A)_{dist}$ = fractional recovery of LK in distillate (i.e., A=LK)

$(FR_B)_{bot}$ = fractional recovery of HK in distillate or bottom (i.e., B=HK)

α_{AB} = the relative volatility

$$\alpha_{AB} = \alpha_{LK-HK} = \alpha_{C3-C4} = \frac{K_{C3}}{K_{C4}} = \frac{1.37}{0.35} = 3.914$$

$$N_{min} = \frac{\ln \left[\frac{0.995 \times 0.99}{(1 - 0.995)(1 - 0.99)} \right]}{\ln 3.914}$$

$$= 7.25 = 8 \text{ stages}$$